



Universidade do Minho
Escola de Engenharia
Departamento de Informática

APRENDIZAGEM AUTOMÁTICA

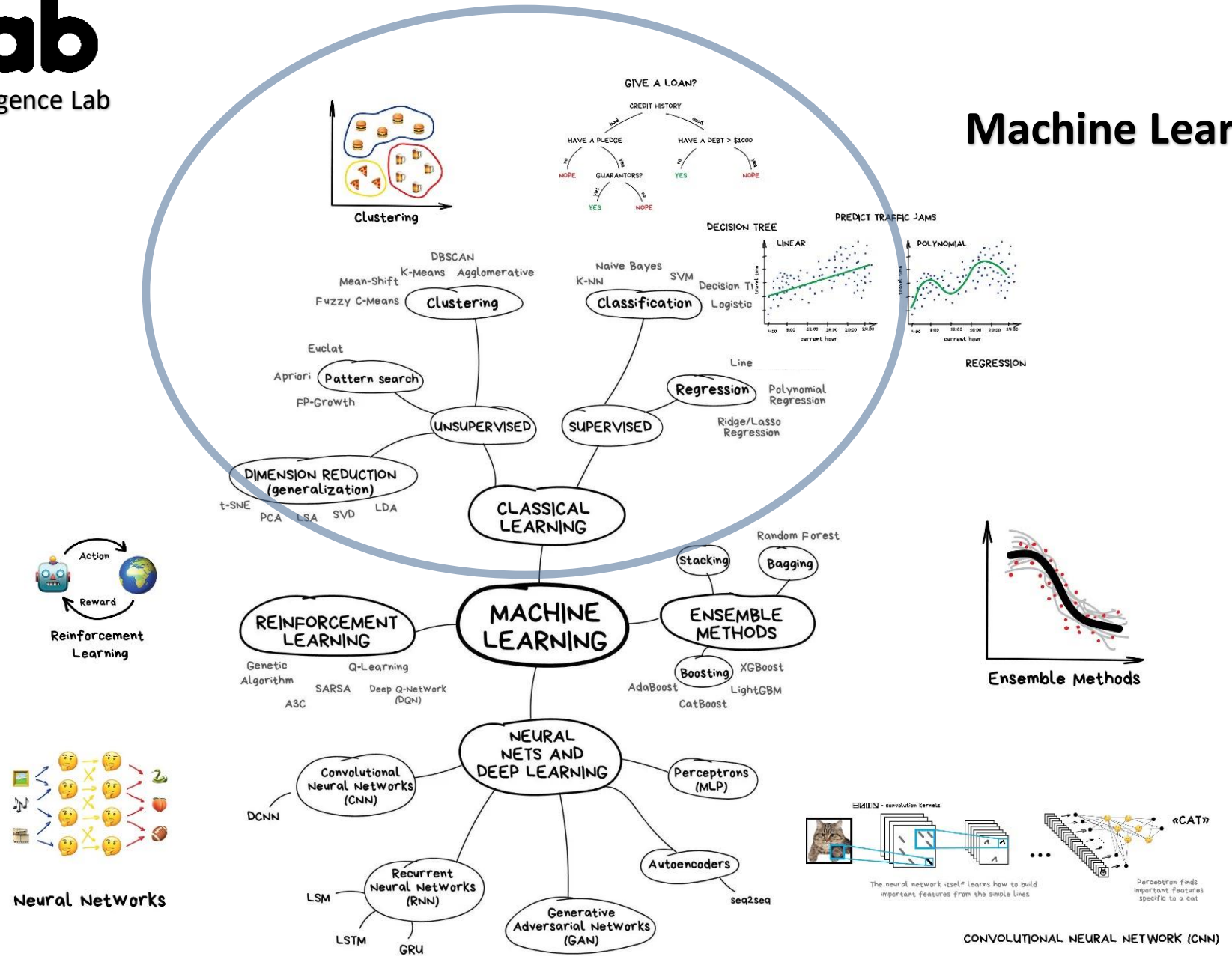
Machine Learning

Model Evaluation and Performance Metrics

Mestrado em Engenharia Telecomunicações e Informática
Inteligência Artificial para as Telecomunicações

2025/26

Machine Learning



- Performance Metrics;
- Model Evaluation;
- Quality metrics;
- Learning Curves in Machine Learning;
- Overfitting and Underfitting;
- Unrepresentative datasets;
- Regression metrics;
- Classification metrics;

- **Performance Metrics** are ways of measuring the quality/utility/validity of a model;
- In most cases they depend on the type of model being evaluated;
- **Regression Metrics;**
 - Used to predict continuous values.
- **Classification Metrics;**
 - Used to predict discrete values.

- Once a model has been created (trained) using a Machine Learning technique, its performance must be evaluated;
- Measuring the performance of a model is done with data not presented during training.

Model Evaluation

Training data:

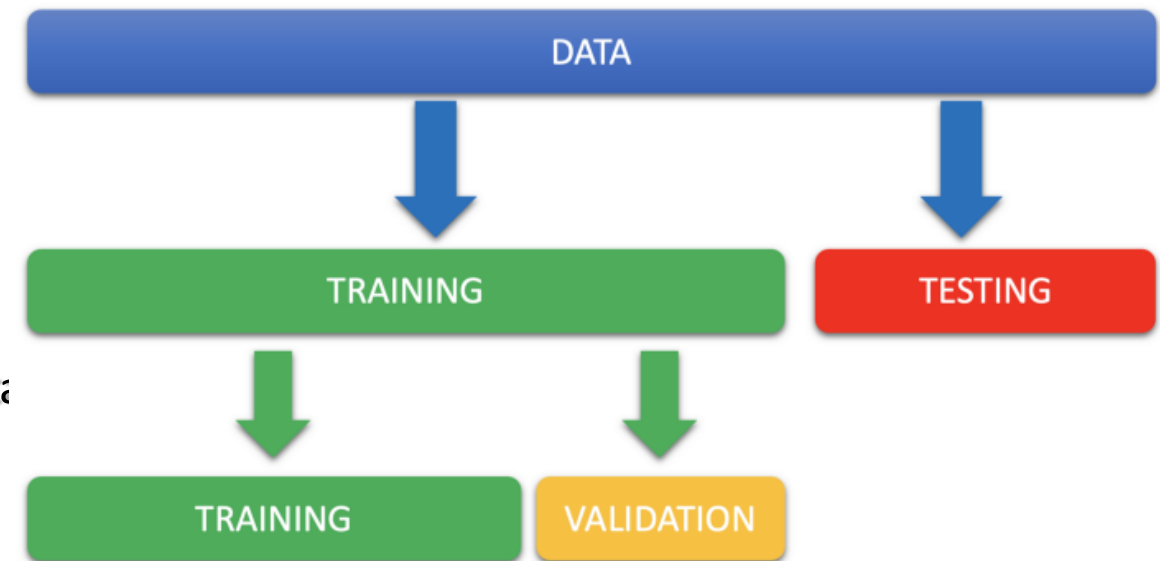
- Dataset used to fit the model;

Validation data:

- Dataset used to provide an unbiased assessment of a model fit, on the training data;

Test data:

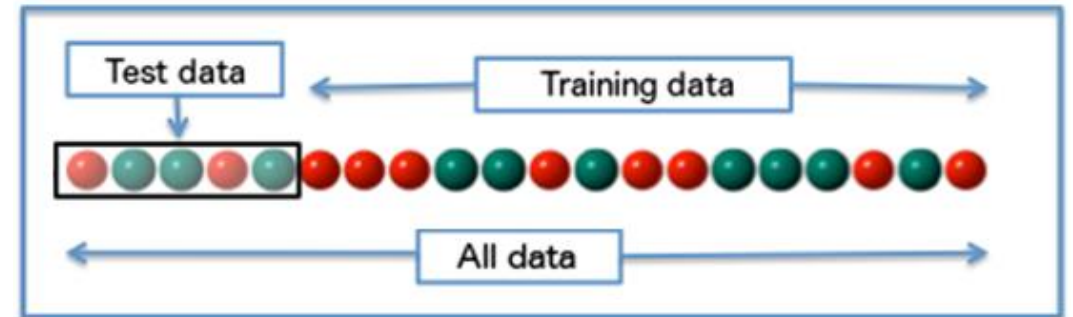
- Dataset used to provide an unbiased assessment of a final model fit to the training data set.



(<https://machinelearningmastery.com/difference-test-validation-datasets>)

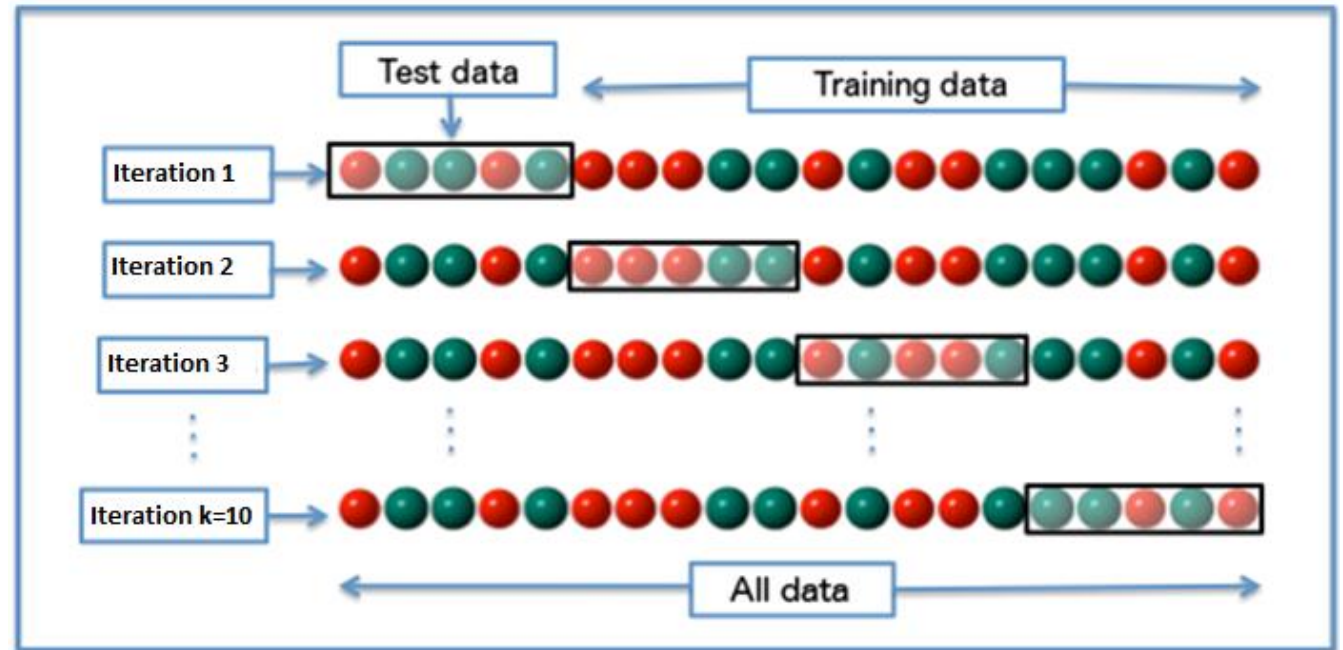
Hold-out Validation

- Data partitioning method;
- Divides the data set into training data and test data;



- A part (hold-out) of the data set is separated for training/testing (80/20; 75/25; ...)

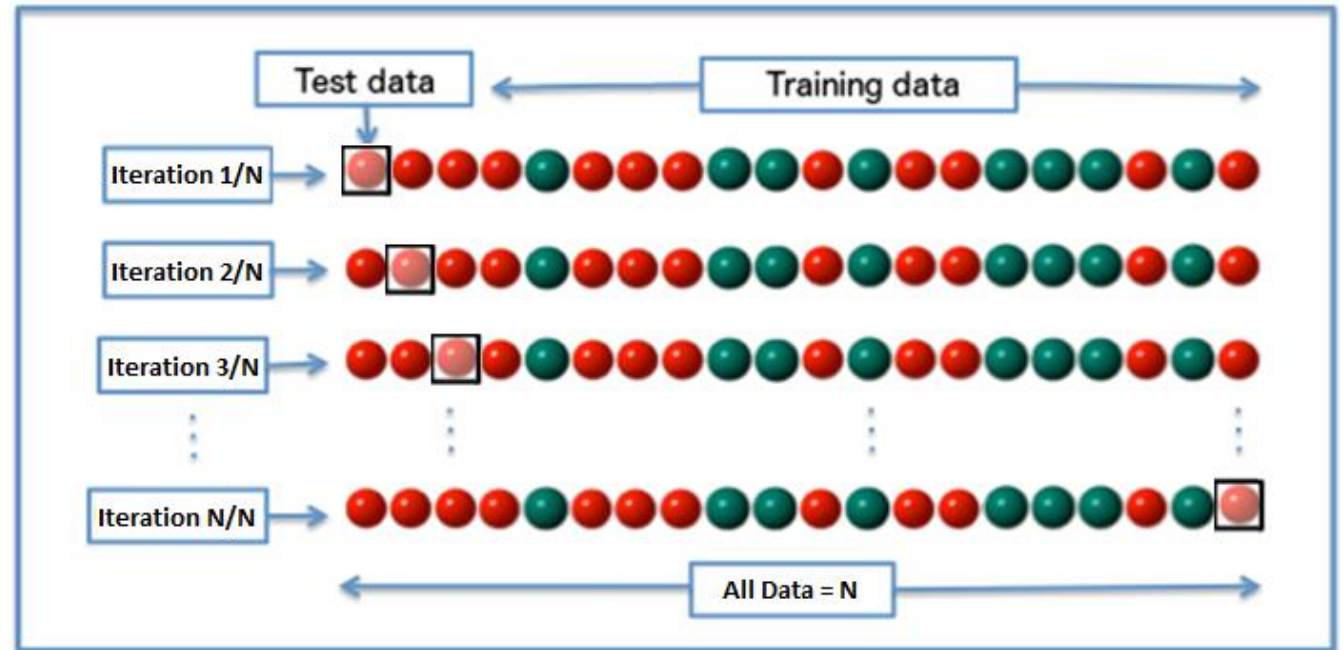
- Cross-validation method;
- Consists of dividing the data set into k parts (k folds);
 - At each iteration, the method uses $k-1$ parts (folds) for training and 1 part (fold) for testing;
 - The process is repeated k times;



- The final error is given by the average of the partial error values.

Leave-one-out Cross Validation ($k=N$)

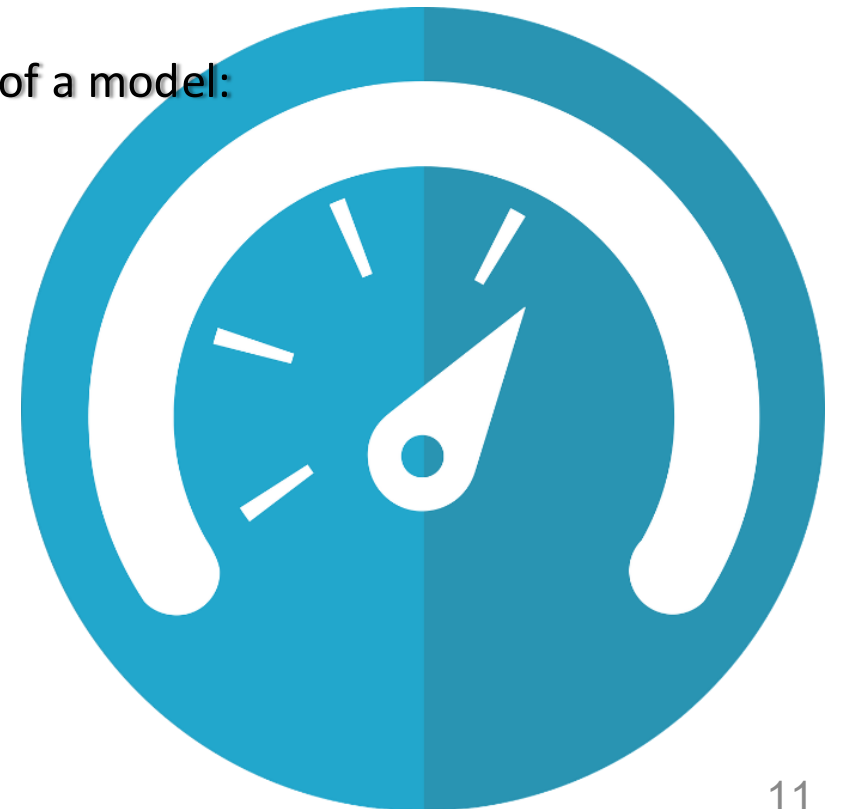
- Cross-validation method;
- Particular case in which the number of cases N is equal to the number of folds k ;



- What is the ideal number for k (folds)?
- If the dataset is large, a small value for k may be sufficient, since we will have a large amount of data for training;
- If the dataset is small, a large value of $k \approx N$ may be more appropriate to maximise the amount of data for training;
- The greater the number of folds, the better the error estimate, the lower the bias(*) and the lower the overfitting;
- In fact, the value of k depends on the value of N !

(*) Bias = distort
= distortion
= prejudice

- Why quality metrics?
 - To evaluate the model's performance.
- Metrics are used to monitor and measure the performance of a model:
 - Mean Absolute Error (MAE),
 - Mean Squared Error (MSE),
 - Precision,
 - F1-Score,
 - among others...
- However, it depends on the problem at hand:
 - Is it a classification problem?
 - Regression?
 - Time series?



Learning Curves in Machine Learning

- They are a plot of the model's (usually) error measure during training, over time/experience
- They are a diagnostic tool that allows you to see how quickly/well the model learns the patterns, and whether there are any problems with the data.
 - Overfitting? Underfitting? Is the data representative?
- Training and validation splits are used

Learning Curves

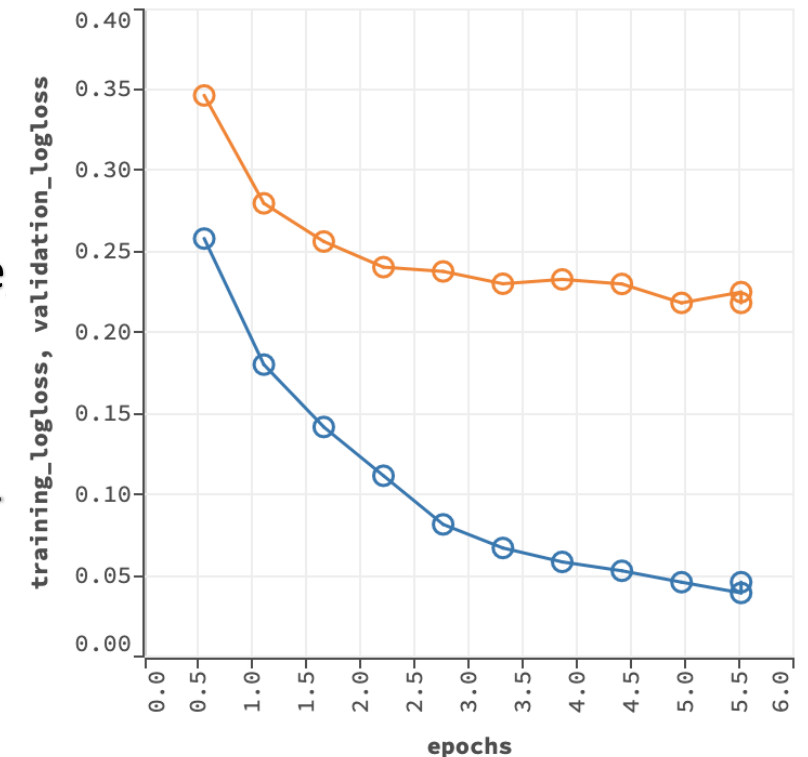
The y-axis can have

- An error measure (to be minimised, e.g. logloss, RMSE) or
- A performance measure (which you want to maximise e.g. accuracy)

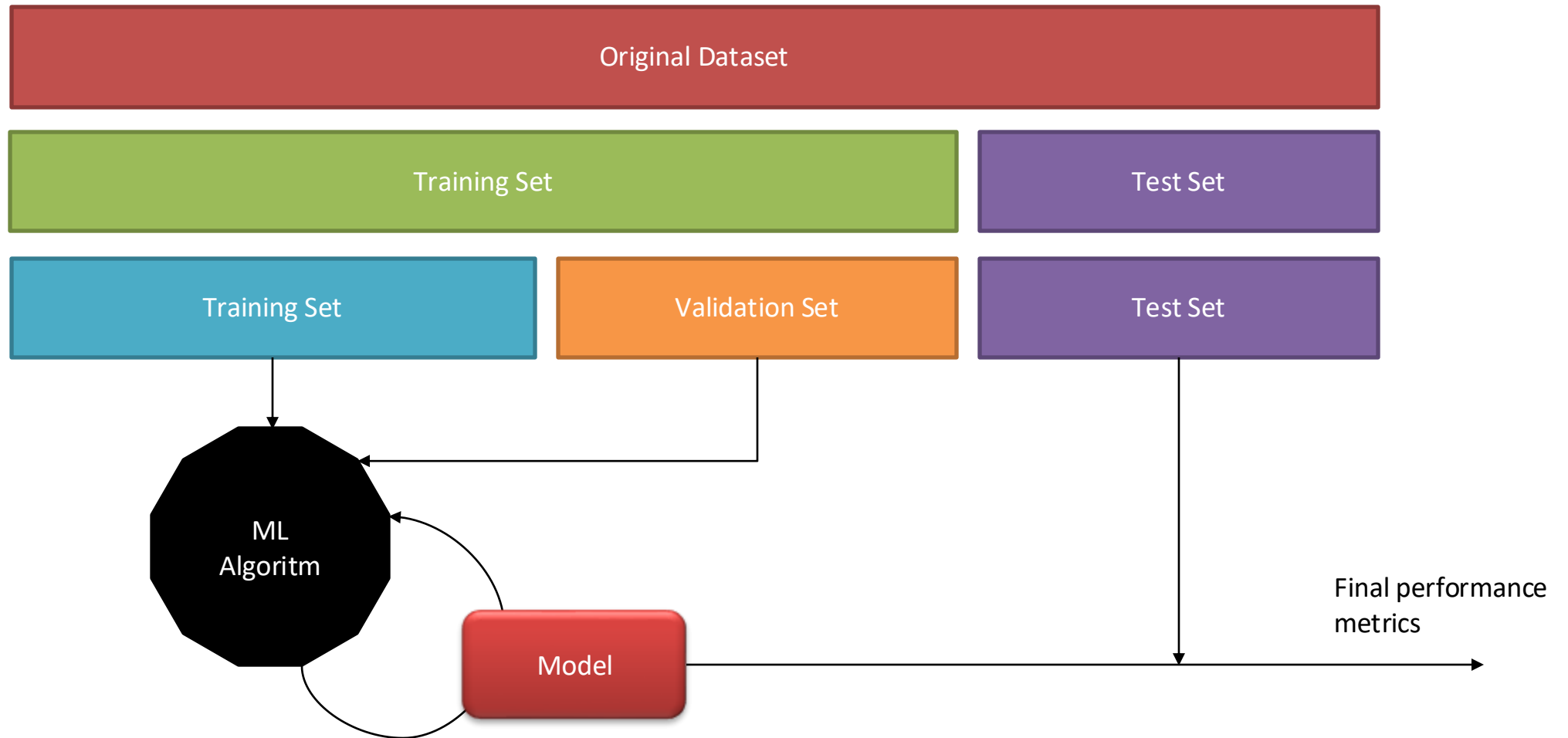
Main idea

- Evaluate model quality throughout training, at regular intervals
- Can be assessed on training and/or validation data (in the case of the latter, it allows generalisation to be assessed)

▼ SCORING HISTORY - LOGLOSS

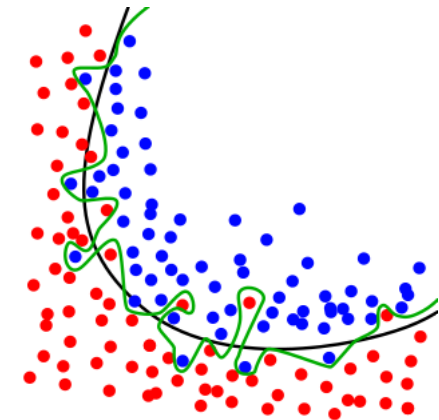
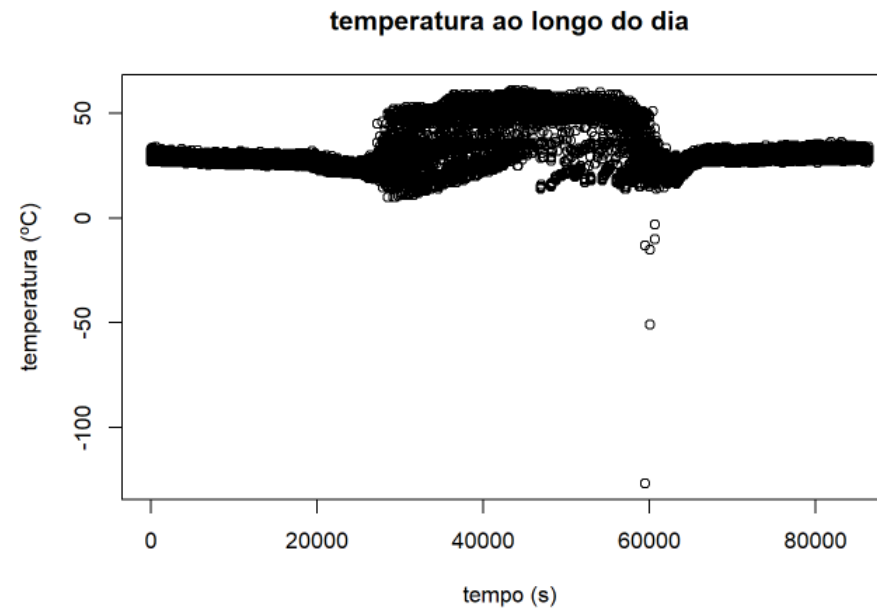
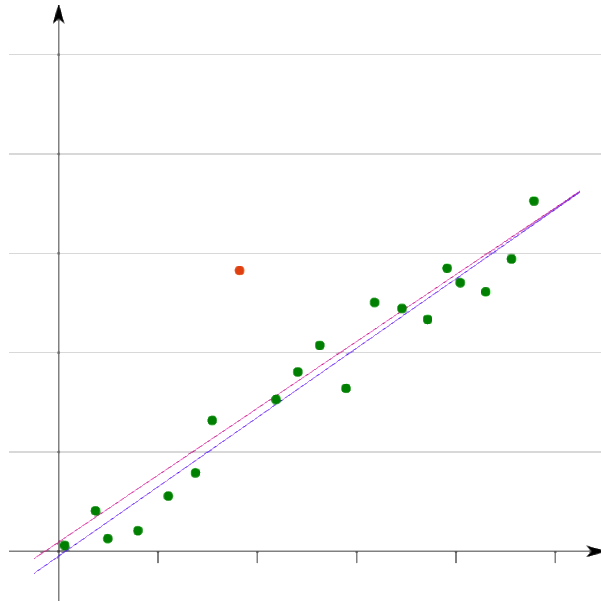


Training, Validation and Test sets



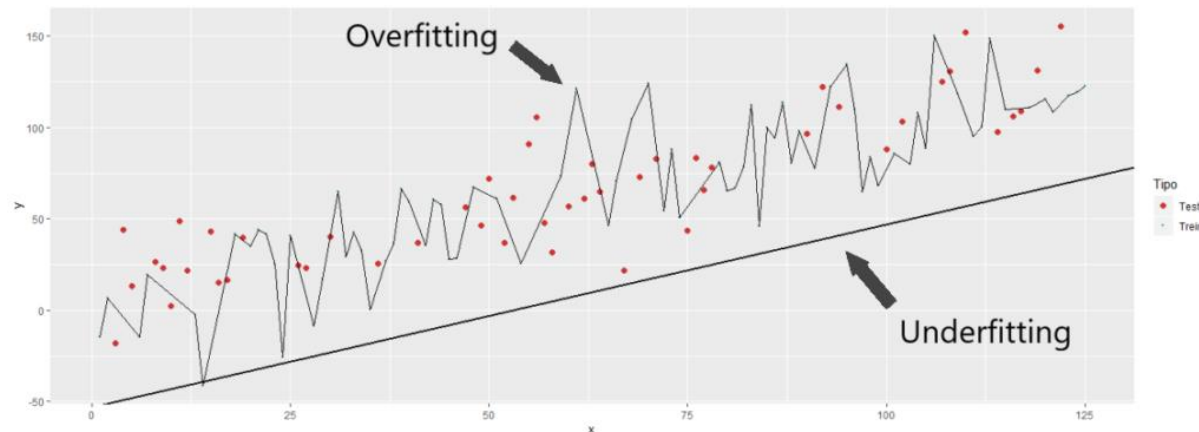
Overfitting

- Signal vs. Noise;
- A good ML algorithm must be able to ignore noise.



- Overfitting happens when the model grows in complexity to fit the data "too well":
 - Predicts better and better on training data;
 - Predicts worse and worse on test/validation data (not known during training);
- Is said to be too sensitive to noise, has a **high variance**;
- Learning is as much about memorising as it is about (knowing) how to forget;
- An algorithm must be able to generalize:
 - Be able to ignore specific (and probably irrelevant) details:
 - Be able to consider only "the big picture", abstract a high-level model.

- Problem opposite to overfitting, model has high bias;
- is a scenario where a data model is unable to capture the relationship between the input and output variables accurately, generating a high error rate on both the training set and unseen data;
- Occurs when the model is too simple to the point of not being able to properly model the data:
 - For example, because it has too few variables.



Source image: <https://didatica.tech/underfitting-e-overfitting/>

Bias vs. variance tradeoff

- Variance

- Refers to the sensitivity of an algorithm to specific sets of data. A model with high variance changes significantly with the addition of some data on the "frontier";

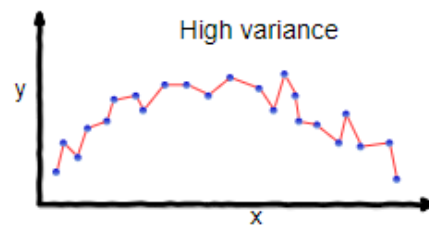
- Bias

- Error that occurs due to oversimplification. Occurs when an algorithm has limited flexibility to learn the true pattern of a dataset;

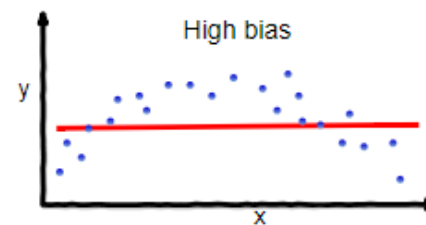
- Generalisation error

- Error measured in the test data;

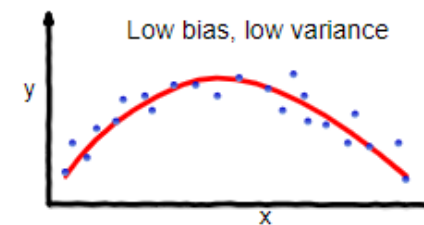
- The goal of an algorithm is to have low bias and variance.



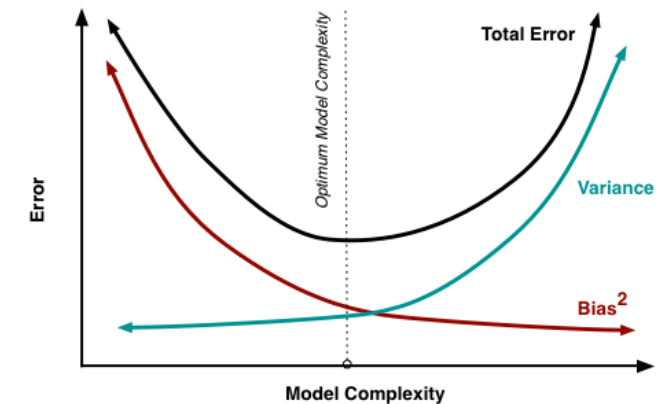
overfitting



underfitting

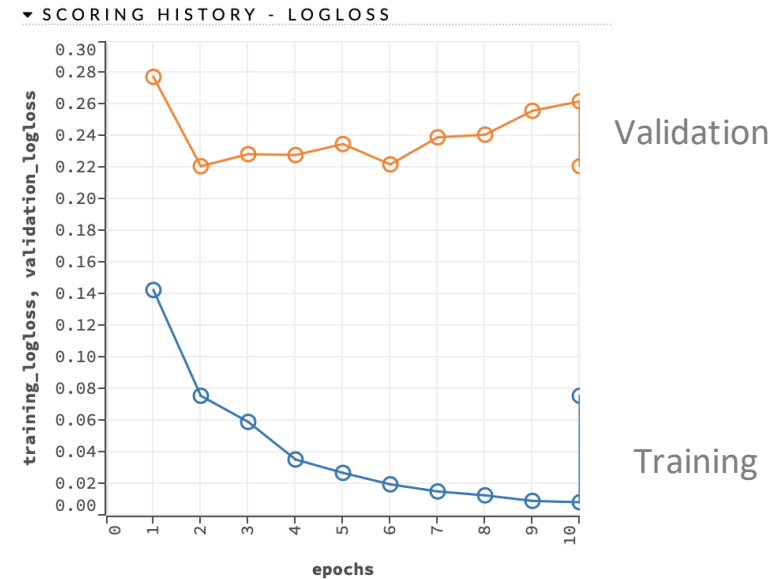
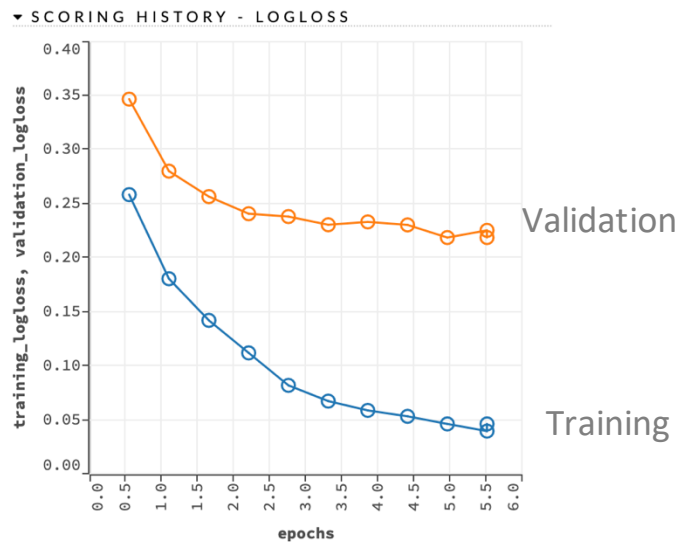


Good balance



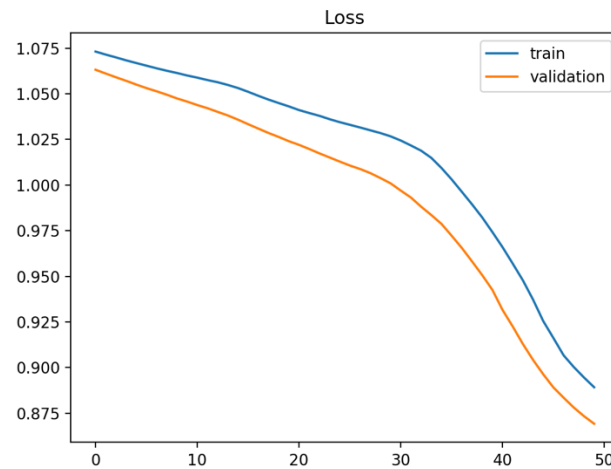
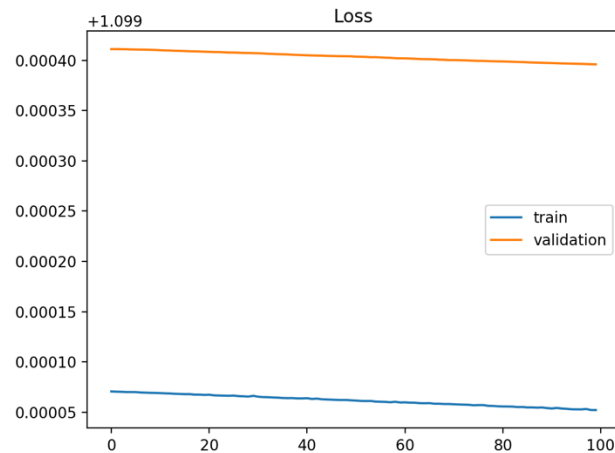
How to detect overfitting

- An overfitting model has
 - Very low training error;
 - A very high validation error or one which, after decreasing, begins to increase.



How to detect underfitting

- An underfitting model has
 - High training error;
 - Validation error similar in magnitude to training error;
 - Error tends not to decrease significantly over time;
 - Or error decreases and is still decreasing significantly in the final stage of training.



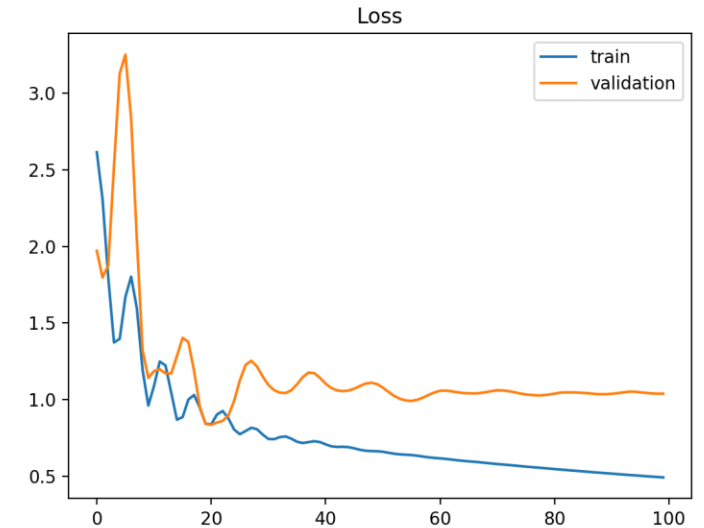
Source: "How to use Learning Curves to Diagnose Machine Learning Model Performance", Jason Brownlee, Deep Learning Performance, 2019.

How to detect unrepresentative datasets

- A dataset is considered unrepresentative when it doesn't properly represent the characteristics of the "real" data, or when compared to another dataset (e.g. training vs. testing);
- It often happens when the dataset is small;
- When training a model, both the training and validation datasets can be unrepresentative:
 - It can be difficult to distinguish between the two situations without knowing the size of the datasets.

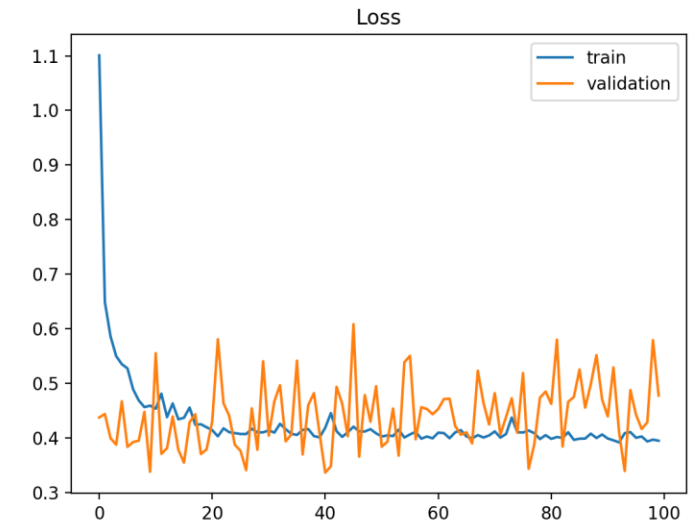
How to detect unrepresentative datasets

- It happens when the training dataset doesn't have enough data to represent the patterns;
- Detected when:
 - There are large variations in the error measure in the initial phase;
 - Training and validation errors decrease but remain distant.



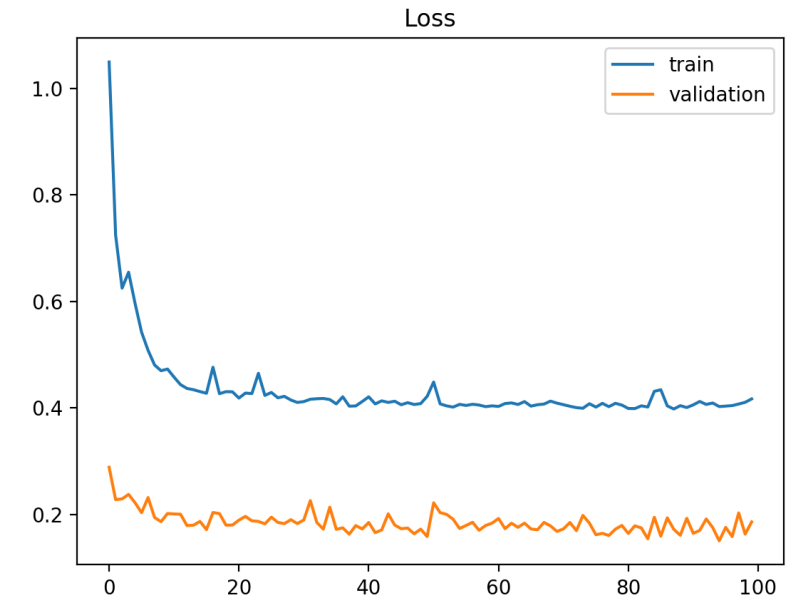
How to detect unrepresentative datasets

- It happens when the validation dataset does not have enough data to assess the model's ability to generalize;
- Detected when:
 - Training error has the expected behavior;
 - Validation error does not converge and has too much variability.



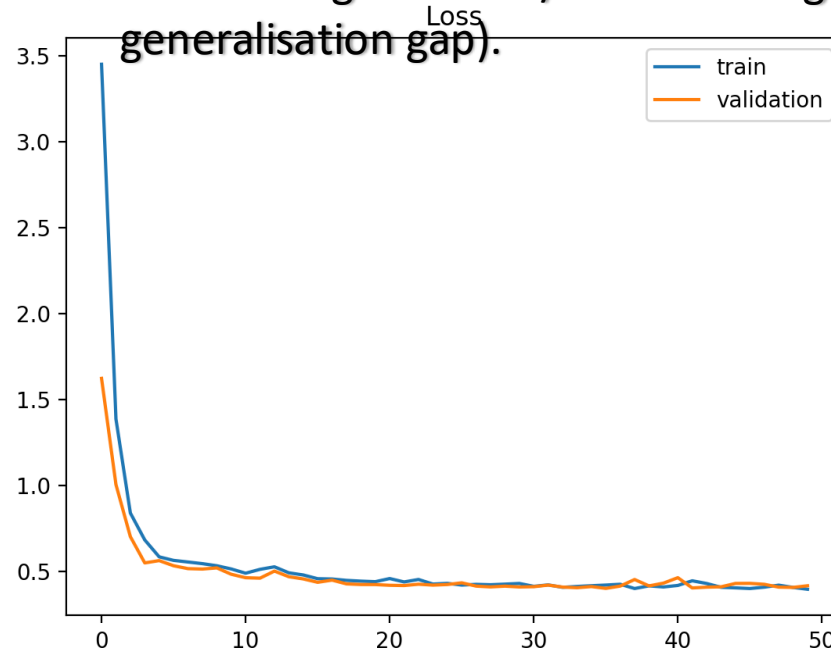
How to detect unrepresentative datasets

- It can also be detected by a lower validation error than the training error, which may indicate "luck" in splitting the data.



How to detect a good model

- In a good model:
 - The training and validation lines are generally close, drop initially, and tend to stagnate towards the end;
 - The training error can/should be slightly lower than the validation error (the difference is called the **generalisation gap**).



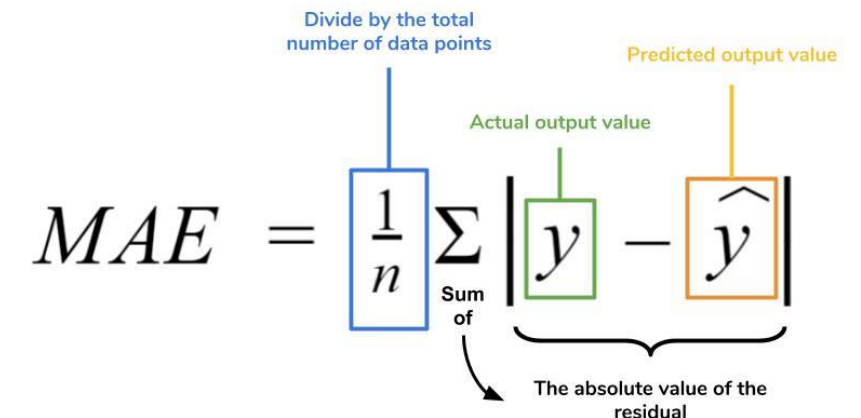
Source: "How to use Learning Curves to Diagnose Machine Learning Model Performance", Jason Brownlee, Deep Learning Performance, 2019.

- **Choosing an appropriate model** (or an Ensemble);
- **Controlling the complexity of the chosen algorithm:**
 - In a decision tree, choosing a greater or lesser depth;
 - In a Random Forest by choosing more/less trees;
 - In a neural network / Deep Learning by increasing/decreasing the number of layers or the number of epochs;
 - ...

■ Mean Absolute Error (MAE)

- Measures the average magnitude of errors in a set of forecasts (does not take direction into account):
- $MAE = \frac{1}{n} \sum_{j=1}^n |y_j - \hat{y}_j|$

where n is the number of observations, y_j e $\hat{y}_j$ are, respectively, the current observation and the predicted value.



The diagram illustrates the MAE formula with color-coded annotations: a blue box around $\frac{1}{n}$ is labeled "Divide by the total number of data points"; a green box around y is labeled "Actual output value"; an orange box around $\hat{y}$ is labeled "Predicted output value"; and a bracket under the absolute value term $|y - \hat{y}|$ is labeled "The absolute value of the residual".

$$MAE = \frac{1}{n} \sum \left| y - \hat{y} \right|$$

- Average value of the differences between observations and predicted values;
- Units are the same as for the response variable.

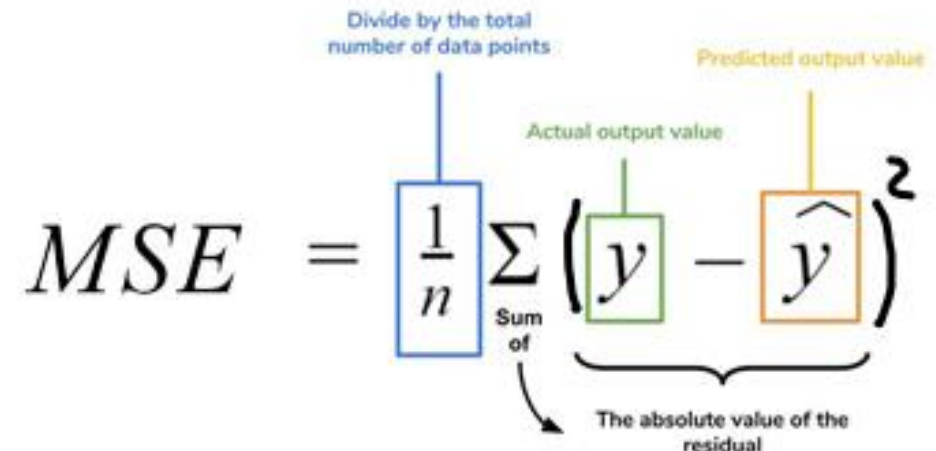
■ Mean Squared Error (MSE)

- It consists of calculating the average of the squared differences between the errors in a set of forecasts (does not take direction into account):

$$\text{MSE} = \frac{1}{n} \sum_{j=1}^n (y_j - \hat{y}_j)^2$$

where n is the number of observations, y_j e $\hat{y}_j$ are, respectively, the current observation and the predicted value.

- Measures the average of the squares of the error
- The greater the error, the greater the "penalty";
- The lower the better (ideally 0);
- Units are the same as for the response variable.



The diagram illustrates the Mean Squared Error (MSE) formula with the following components and annotations:

- Divide by the total number of data points:** Points to the $\frac{1}{n}$ term.
- Sum of:** Points to the summation symbol $\sum$.
- Actual output value:** Points to the y term inside the square.
- Predicted output value:** Points to the $\hat{y}$ term inside the square.
- The absolute value of the residual:** Points to the difference $y - \hat{y}$ inside the square.
- Squared:** Points to the exponent 2 .

$$MSE = \frac{1}{n} \sum (y - \hat{y})^2$$

- Root Mean Squared Error (RMSE)

- Calculates the average of the squared differences between the errors in a set of forecasts (does not take direction into account):

- $RMSE = \frac{1}{n} \sqrt{\sum_{j=1}^n (y_j - \hat{y}_j)^2}$

where n is the number of observations, y_j e $\hat{y}_j$ are, respectively, the current observation and the predicted value.

$$RMSE = \sqrt{\sum_{i=1}^n \frac{(\hat{y}_i - y_i)^2}{n}}$$

- Represents the standard deviation of predicted values from observed values;
- Like other measures, it depends on the scale, so it can't be used to compare different datasets (only to compare different models for the same dataset);
- Units are the same as for the response variable.

Regression Metrics

■ R² Score

- Quantifies the percentage of the variation in the dependent variable that is explained by the independent variables;
- Varies between 0 and 1 (the higher, the better);
- SSt - Sum of the distance from each point to the mean value (variance of the test data);
- SSr - Sum of the distance of each point from the value predicted by the model (model variance);
- determines the proportion of variance in the dependent variable that can be explained by the independent variable. In other words, r-squared shows how well the data fit the regression model (the goodness of fit).

$$SS_t = \sum_{i=1}^m (y_i - \bar{y})^2$$

$$SS_r = \sum_{i=1}^m (y_i - \hat{y}_i)^2$$

$$R^2 \equiv 1 - \frac{SS_r}{SS_t}$$

Regression Metrics

$$MAE = \frac{1}{n} \sum_{j=1}^n |y_j - \hat{y}_j| \quad MSE = \frac{1}{n} \sum_{j=1}^n (y_j - \hat{y}_j)^2 \quad RMSE = \frac{1}{n} \sqrt{\sum_{j=1}^n (y_j - \hat{y}_j)^2}$$

- Three of the most common metrics used to measure the accuracy of continuous variables;
- All express the average prediction error of the model (lower values are better);
- All range from 0 to ∞ and are indifferent to the direction of the errors;
- MAE and RMSE express the prediction error in the same unit as the variable of interest;
- MSE and RMSE, by squaring the error, give relatively high weight to large errors;
- MSE and RMSE are more useful when large errors are particularly undesirable.

Regression Metrics

$$MAE = \frac{1}{n} \sum_{j=1}^n |y_j - \hat{y}_j|$$

$$MSE = \frac{1}{n} \sum_{j=1}^n (y_j - \hat{y}_j)^2$$

$$RMSE = \frac{1}{n} \sqrt{\sum_{j=1}^n (y_j - \hat{y}_j)^2}$$

#	Error	Error	Error ²
1	1	1	1
2	-1	1	1
3	3	3	9
4	3	3	9

**MAE MSE RM
SE**

2 5 2.24

#	Error	Error	Error ²
1	0	0	0
2	0	0	0
3	0	0	0
4	10	10	100

**MAE MSE RM
SE**

2.5 25 5

Classification metrics

■ Confusion Matrices

- Table used to describe the performance of a classification model.
- It shows how many instances were correctly classified and also which classes were chosen when errors occurred;
- Information can be used to try to improve the model;
- Allows you to assign a degree of confidence (or lack thereof) to the predictions.

		True Class	
		Positive	Negative
Predicted Class	Positive	TP	FP
	Negative	FN	TN

Classification metrics

- Confusion Matrices

- Table used to describe the performance of a classification model.

- *Accuracy*

- Number of correct predictions divided by the total number of observations:

- $$Accuracy = \frac{TP+TN}{TP+TN+FP+FN}$$

Proportion of correct classifications in relation to the total number of instances

"In general, how many cases does the model get right?";

Total number of correct classifications divided by total number of instances.

		True Class	
		Positive	Negative
Predicted Class	Positive	TP	FP
	Negative	FN	TN

■ Confusion Matrices

- Table used to describe the performance of a classification model.

■ Precision (*aka Sensitivity*)

- It is a measure of accuracy;
- It determines the proportion of relevant items among all items:

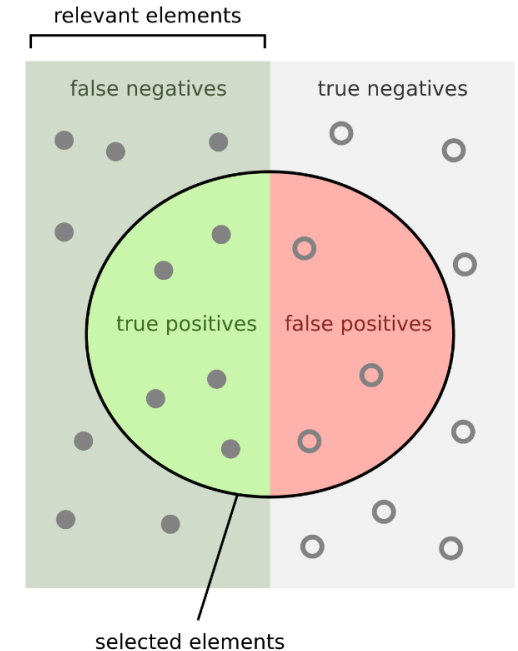
$$\bullet \text{ Precision} = \frac{TP}{TP+FP}$$

Proportion of correct positive predictions related to cases predicted as positive;

"Of the cases classified as positive, how many are actually positive?";

Number of well-classified positives divided by the total number of predicted positives (right

Classification metrics



How many selected items are relevant?

$$\text{Precision} = \frac{\text{Green}}{\text{Green} + \text{Red}}$$

How many relevant items are selected?

$$\text{Recall} = \frac{\text{Green}}{\text{Green} + \text{Gray}}$$

■ Confusion Matrices

- Table used to describe the performance of a classification model.

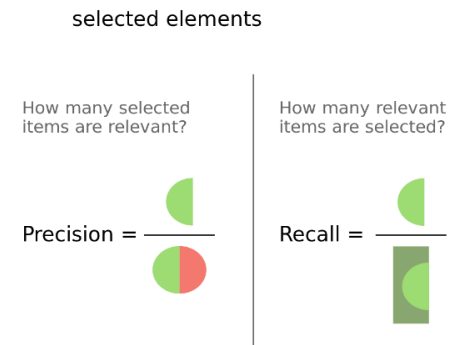
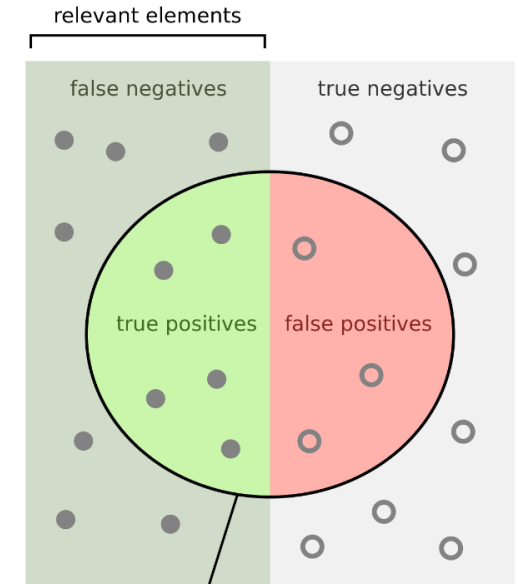
■ *Recall (aka Specificity)*

- It is a measure of completeness;
- It determines the proportion of relevant items obtained:

$$\bullet \text{ Recall} = \frac{TP}{TP+FN}$$

- Number of correctly classified positives divided by the actual number of positives;
- "What percentage of positives did the model find?"

Classification metrics



■ Confusion Matrices

- Table used to describe the performance of a classification model.

■ Precision (*aka Sensitivity*)

- It is a measure of accuracy;
- It determines the proportion of relevant items among all items:

$$\bullet \text{ Precision} = \frac{TP}{TP+FP}$$

■ Recall (*aka Specificity*)

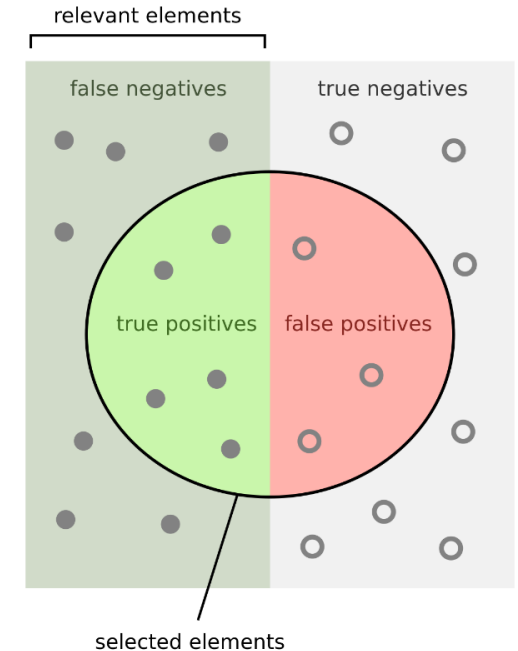
- It is a measure of completeness;
- It determines the proportion of relevant items obtained:

$$\bullet \text{ Recall} = \frac{TP}{TP+FN}$$

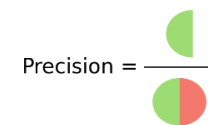
Ideally, precision and recall should be high;

In practice, we end up erring more in one direction or the other (i.e. if recall goes up, precision goes down and vice versa).

Classification metrics

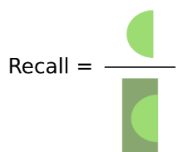


How many selected items are relevant?



$$\text{Precision} = \frac{\text{Green}}{\text{Green} + \text{Red}}$$

How many relevant items are selected?



$$\text{Recall} = \frac{\text{Green}}{\text{Green} + \text{Light Gray}}$$

Classification metrics

■ Confusion Matrices

- Table used to describe the performance of a classification model.

■ Precision (*aka Sensitivity*)

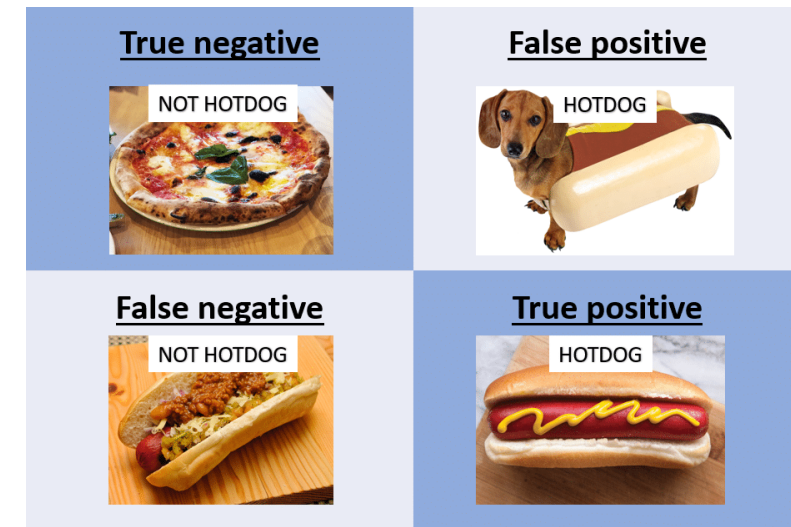
- It is a measure of accuracy;
- It determines the proportion of relevant items among all items:

$$\bullet \text{ Precision} = \frac{TP}{TP+FP}$$

■ Recall (*aka Specificity*)

- It is a measure of completeness;
- It determines the proportion of relevant items obtained:

$$\bullet \text{ Recall} = \frac{TP}{TP+FN}$$



Source image: <https://blog.nillsf.com/index.php/2020/05/23/confusion-matrix-accuracy-recall-precision-false-positive-rate-and-f-scores-explained/>

Binomial classification

Example: covid-19 occurrence

Prevision	Reality	
	Yes	No
Yes	TP 1770	FP 150
No	FN 330	TN 97750

Binomial classification

Example: covid-19 occurrence

Prevision	Reality	
	Yes	No
Yes	TP 1770	FP 150
No	FN 330	TN 97750

Patients who do not have covid but whom the model predicts will have it

Patients who have covid but the model predicts they don't

Binomial classification

Prevision	Reality	Yes	No
Yes		TP 1770	FP 150
No		FN 330	TN 97750

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} = \frac{1770+97750}{100000} = 0,9952$$

← In general, the model is correct in 99.52 per cent of cases

$$Precision = \frac{TP}{TP+FP} = \frac{1770}{1770+150} = 0,9219$$

← When the model predicts that you have covid, the patient has an 8% chance of not having covid.

$$Recall = \frac{TP}{TP+FN} = \frac{1770}{1770+330} = 0,8428$$

← The model finds 84.28 per cent of the cases of covid

Classification metrics

■ Confusion Matrices

- Table used to describe the performance of a classification model.

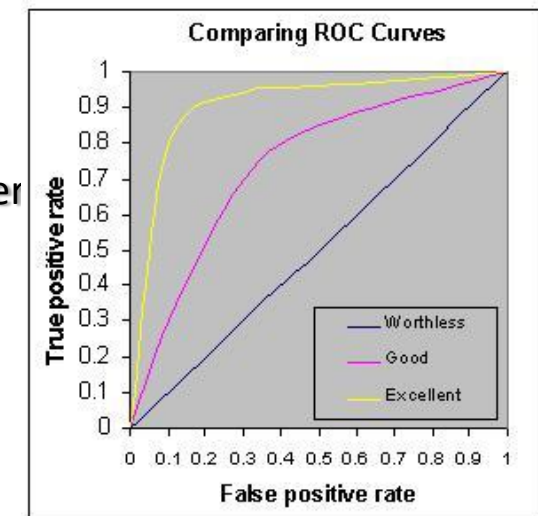
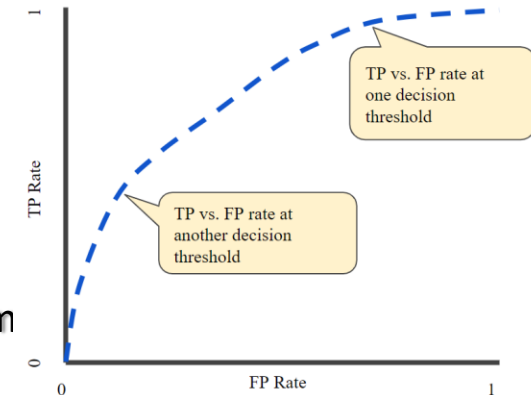
■ ROC curve:

- The Receiver Operating Characteristics (ROC) curve finds the performance of a classification model at different classification thresholds;
- By reducing the classification threshold, more items are classified as positive, increasing false positives and true positives.

Graphical representation illustrating the performance of a binary classifier

X axis -> FP

Y axis -> TP



- Confusion Matrices

- Table used to describe the performance of a classification model.

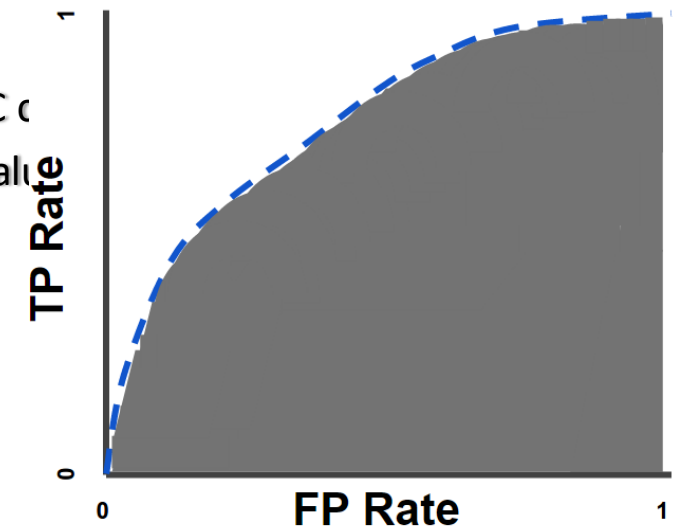
- AUC curve:

- The Area Under the Curve (AUC) measures the area under the ROC curve.
 - It measures how well the predictions are classified, rather than evaluating their absolute values (ranges from 0 to 1);
 - A model whose predictions are 100% wrong has an AUC of 0; one whose predictions are 100% correct has an AUC of 1.

Generally varies between 0.5 and 1;

The higher the better;

If close to the diagonal, classifier performs similarly to random selection.



See also: <http://www.navan.name/roc/>

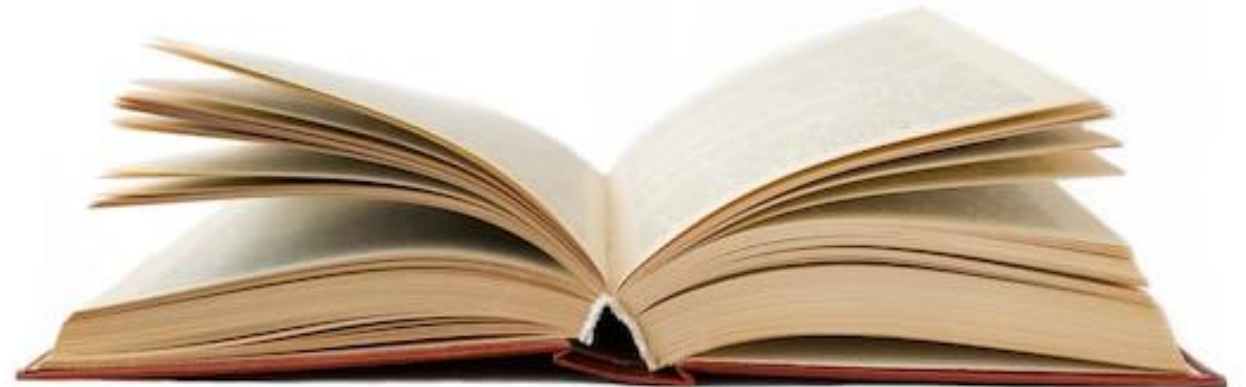
F-measure (F-score)

- “Absolute” measure of performance, combining Precision with Recall
 - There are cases in which it is preferable to maximise precision;
 - There are cases in which it is preferable to maximise recall.

$$F_{\beta} = \frac{1}{\beta * \frac{1}{Precision} + (1 - \beta) * \frac{1}{Recall}}$$

- The higher the β , the greater the importance given to Precision;
- The highest possible value of an F-score is 1.0, indicating perfect precision and recall, and the lowest possible value is 0, if either precision or recall are zero.

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APRENDIZAGEM AUTOMÁTICA

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